CE 310 — Week 1 Assignment: Why Averages Lie

Due: Wednesday of Week 2 by 9:00 AM — submit to D2L

Overview

This assignment builds directly on the Week 1 lab. You will extend your analysis beyond temperature and make a concrete design recommendation — the kind of decision a practicing engineer would write in a project report.

How to Submit

> ☐ Following this submission format exactly is essential — with a class this size, grading depends on it. Submissions that don't comply will be penalized.

Requirement	Detail
File format	.xlsx only — written responses go in the Answer Sheet cells, not in a separate PDF
Sheet name	Must be exactly Answers — do not rename or delete it
Major declaration	Enter CE or ArcE in the Major cell at the top of the sheet — this determines which of the 2b/2c cells is graded
Column layout	Enter answers only in the labeled input cells for each problem — column position varies by section (see each section's own column headers). Do not insert, delete, or move columns
Formulas	Type them as text — put an apostrophe before the =, e.g. '=COUNTIF(B2:B8761,">90"), so Excel stores the formula instead of running it. Screenshots are not accepted
File name	CE310_W1_<NetID>.xlsx e.g. CE310_W1_abc123.xlsx
Submission	Upload the .xlsx file to D2L Dropbox before the deadline

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Problem 1 — Exceedance Hours (30 pts)

Using Week1_Tucson_Weather_2023.csv, answer the following.

1a. How many hours in 2023 did Tucson's temperature exceed each threshold below? Complete the table.

Threshold (°F)	Hours Exceeding	% of Year
85		
90		
105		

> Your lab questions A6 and A7 already computed the counts for >95 °F and >100 °F — you do not need to repeat those here.

Use COUNTIF for each row. Type each formula as text in Col D (Formula Used) of your Answer Sheet — screenshots are not accepted.

1b. Using your ASHRAE 1% design dry-bulb temperature from the lab (the 88th-highest hourly temperature, found with =LARGE(B2:B8761, 88)), calculate the peak cooling load difference between two design approaches for a 10,000 sq ft office building:

- Approach A (wrong): Size the system for the annual average temperature.
- Approach B (correct): Size the system for the ASHRAE 1% design temperature.

The simplified sensible cooling load formula is:

$Q = U \times A \times \Delta T$

where Q = cooling load (BTU/hr), U = 0.5 BTU/(hr·ft²·°F) (overall heat transfer coefficient), A = 10,000 ft² (floor area), and ΔT = indoor setpoint (75 °F) minus outdoor design temperature.

Calculate Q for both approaches and find the difference. In 2–3 sentences, explain what it means in practice if the installed system can only handle Approach A's load on a peak summer day.

Enter your response in the P1b_written cell of your Answer Sheet.

1c. Using B2 from your lab (the gap between the ASHRAE design temperature and the annual average) and A7 (hours above 100 °F), write 2–3 sentences explaining what a building occupant would experience during a peak summer day if the HVAC system was sized for the annual average temperature. Focus on comfort — when does the system fall short, and for how many hours of the year?

Enter your response in the P1c_written cell of your Answer Sheet.

Problem 2 — Seasonal Humidity Analysis and Ventilation Interpretation (18 pts)

Your lab computed the annual average relative humidity (B12) and hours above 60% RH (B13). Now compute the seasonal breakdown — this is new data not in your lab.

2a. Calculate the average relative humidity for: - Winter: December, January, February (months 12, 1, 2) - Summer: June, July, August (months 6, 7, 8)

Use AVERAGEIF with the Month column (one call per month, then average the three results), or AVERAGEIFS if your Excel version supports it.

2b/2c — answer ONLY the prompt matching your major. Declare your major (CE or ArcE) at the top of your Answer Sheet — that declaration determines which of the two cells below is graded. Leave the other one blank.

2b (Civil Engineering). The difference between winter and summer RH is substantial. Tucson's North American Monsoon runs July–September, driven by Gulf of California moisture. In 3–4 sentences: explain what this seasonal humidity swing means for HVAC latent load sizing compared to a design based only on the annual average, and what the consequence is for equipment capacity if it's ignored.

CE students: enter your response in the P2_CE_written cell of your Answer Sheet.

2c (Architectural Engineering). ASHRAE 62.1-2022 requires that the outdoor air quantity delivered to a space be based on occupancy rate and floor area. In 3–4 sentences: explain why using annual-average relative humidity rather than peak-week (monsoon) relative humidity would underestimate the latent cooling load on a mechanical ventilation system — and what the consequences are for occupant comfort and indoor air quality during the monsoon season.

ArcE students: enter your response in the P2_ArcE_written cell of your Answer Sheet.

Problem 3 — Reflection (12 pts)

Answer in no more than one paragraph (100–150 words):

A colleague argues: "We should always design for the average case to keep costs reasonable — extreme events are rare and unpredictable anyway."

Using at least one specific number from your analysis (an exceedance count, a peak value, or a pressure ratio), write a professional response that either agrees, disagrees, or qualifies this statement. Mention one real consequence (cost, safety, or occupant impact) of choosing the wrong design basis.

Enter your response in the P3_written cell of your Answer Sheet.

Submission Requirements

- Type all written responses in the designated yellow cells of your Assignment Answer Sheet.
- Declare your major (CE or ArcE) at the top of the sheet, and answer only the 2b/2c prompt that matches it.
- Submit CE310_W1_<NetID>.xlsx to D2L.

Grading Summary

Problem	Points
Problem 1a — 3 exceedance thresholds with formulas (3 pts each)	9
Problem 1b — Cooling load comparison + written explanation	12
Problem 1c — Written explanation of comfort impact	9
Problem 2 — Seasonal Humidity Analysis and Ventilation Interpretation	18
Problem 3 — Reflection paragraph	12
Total	60 pts

This assignment is worth 60 raw points and will be scaled to the Post-Lab Memo category in the course gradebook.